

Real Analysis: 1.3

Theresa Wunderlich

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1 1.3 Order and Absolute Value

Although the real numbers satisfy the field axioms given in the previous section, these algebraic properties don't fully characterize $\mathbb{R}$; one thing that's missing is the notion of *order*. We begin by introducing the concept of positive and negative numbers.

Axiom 1.4 (Order). There exists a subset of the real numbers called the set of **positive numbers**, denoted $\mathbb{R}^+$, that satisfies the following two conditions:

- (a) If $a, b \in \mathbb{R}^+$, then $a + b \in \mathbb{R}^+$ and $a \cdot b \in \mathbb{R}^+$. In other words, the positive numbers are closed under addition and multiplication.
- (b) Given any real number a , one and only one of the following is true:
 - (i) a is positive;
 - (ii) a is zero;
 - (iii) $-a$ is positive.

The complement of the set $\mathbb{R}^+ \cup \{0\}$, i.e. the set of numbers a that are neither positive nor zero, is called the set of **negative numbers** and is denoted $\mathbb{R}^-$.

Problem 1. Use the order axioms to verify the following facts.

- (a) Let a be a real number. Show that if a is positive, then $-a$ is negative. Conversely, if a is negative, show that $-a$ is positive.

Proof. Assume that a is a positive number. By part (b) of the axiom, exactly one of the statements must be true, making the other two false. Since a is assumed to be positive, case (i) (a is positive) holds, and the one-and-only-one clause forces the other two cases to fail. Case (ii) failing gives $a \neq 0$; and if $-a = 0$, then adding a to both sides gives $0 = a$, contradicting $a \neq 0$, so $-a \neq 0$. Case (iii) failing means “ $-a$ is positive” is false, so $-a \notin \mathbb{R}^+$. Hence $-a \notin \mathbb{R}^+ \cup \{0\}$, so $-a$ must be negative.

Conversely, suppose a is negative, so $a \notin \mathbb{R}^+ \cup \{0\}$. Then a is not positive (case (i) fails) and $a \neq 0$ (case (ii) fails); the first two cases fail for a . Since one and only one alternative must hold, the third must be true: $-a$ is positive. $\square$

Problem 2. (b) Show that the real number 1 is positive.

Proof. Suppose, for contradiction, that 1 is not positive. Since $1 \neq 0$ (by field axiom (iii): there exists a real number distinct from 0, which we call 1), Axiom 1.4(b) applied to 1 rules out the first two cases ($1 \notin \mathbb{R}^+ \cup \{0\}$) and leaves only the case where -1 is positive. By closure under

multiplication (Axiom 1.4(a)), $(-1) \cdot (-1) \in \mathbb{R}^+$, and since $(-1) \cdot (-1) = 1$, it follows that 1 is positive. But by the result of part (a), -1 being positive makes $1 = -(-1)$ negative. So 1 would be both positive and negative, which Axiom 1.4(b) forbids. Therefore 1 is positive. $\square$

Problem 3. (c) Show that there exists a negative number.

Proof. By part (b), 1 is positive. By part (a), if a number is positive then its negation is negative; taking that number to be 1, we conclude that -1 is negative. Therefore a negative number exists, namely -1 . $\square$